Example

Minimize $0.5x^2 + 3x + 4y$

$\frac{1}{2}[x^T P x] + q^T x$

Subject to

$\mathbf{Gx \leq h}$

$x \geq 0$

$y \geq 0$

$x + 3y \geq 15$

$2x + 5y \leq 100$

$3x + 4y \leq 80$